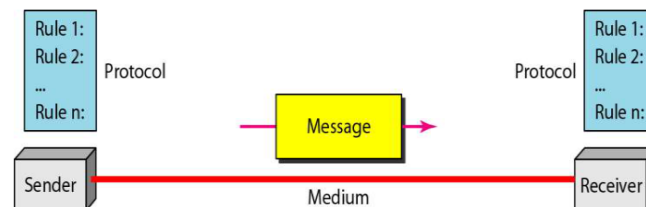


COMPUTER NETWORK AND COMMUNICATION

UNIT - I IMPORTANT QUESTIONS & ANSWERS

Components of a data communication system

1. Message
2. Sender
3. Receiver
4. Transmission medium
5. Protocol



10

Prepared from the uploaded Unit-I PPT, keeping the PPT's terminology, point-wise structure, formulas and visual style as closely as practical.

Handwritten important topics covered

- Data communication, characteristics and components
- Data flow, connections, physical topologies and network types
- Connection-oriented vs connectionless service
- Network devices and network models
- Analog/digital signals, impairment and data-rate limits
- Performance and line coding
- Transmission media and packet switching
- Data-link sublayers, framing, errors, redundancy, parity and CRC

Study strategy: Learn the definitions, the bold points, the formulas, and the diagrams marked in the PPT. The answers are intentionally exam-oriented rather than textbook-long.

1. What is Data Communication? State its characteristics.

Definition: Data communications are the exchange of data between two devices through some form of transmission medium such as a wire cable.

Characteristics

- Delivery: Data should reach the correct destination.
- Accuracy: Data should be delivered without errors.
- Timeliness: Data should reach the destination within the required time.
- Jitter: Variation in packet arrival time; it matters especially for time-sensitive audio and video.

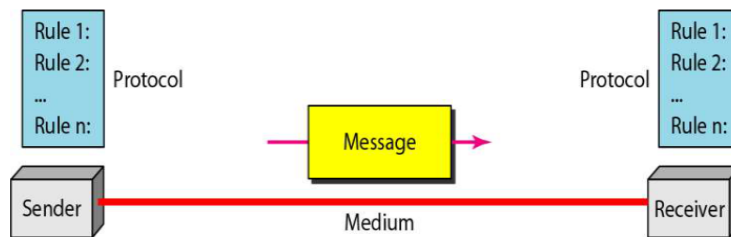
Exam line: A good data communication system provides delivery, accuracy, timeliness and acceptable jitter.

PPT reference: Slides 9-10; jitter is also explained in the Performance section.

2. Explain the Components of a Data Communication System.

Components of a data communication system

1. Message
2. Sender
3. Receiver
4. Transmission medium
5. Protocol



10

Five components

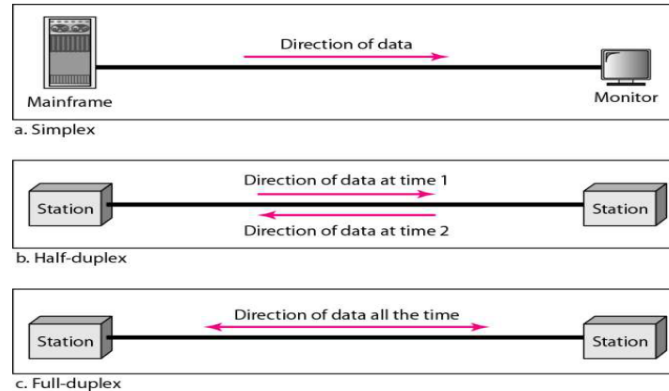
- Message: Information to be communicated; it may be text, numbers, images, audio or video.
- Sender: Device that sends the message.
- Receiver: Device that receives the message.
- Transmission medium: Physical path through which the message travels.
- Protocol: Set of rules governing communication between the devices.

Diagram to reproduce: Sender → Medium → Receiver; Message travels over the medium and Protocol defines the rules.

PPT reference: Slide 10 and the components diagram.

3. Explain Data Flow: Simplex, Half-Duplex and Full-Duplex.

Data flow (simplex, half-duplex, and full-duplex)



12
2

Mode	Direction	Meaning
Simplex	One direction	Data travels only from sender to receiver.
Half-duplex	Both directions, not at the same time	Either device can transmit, but only one transmits at a time.
Full-duplex	Both directions simultaneously	Both devices can transmit at the same time.

PPT reference: Slide 12.

4. What is a Network?

Definition: A network is the interconnection of a set of devices capable of communication.

A computer network connects two or more computing devices for transmitting and sharing information. Devices may use physical wires such as fiber optics or wireless links.

Network criteria

- Performance: Transit time, response time, throughput and delay.
- Reliability: Accuracy of delivery, frequency of failure, recovery time and robustness.
- Security: Protection from unauthorized access/damage and recovery procedures.

Exam line: A network connects devices so information can be transmitted/shared; performance, reliability and security are key criteria.

PPT reference: Slides 13–15.

5. Explain Types of Connections and Physical Topologies.

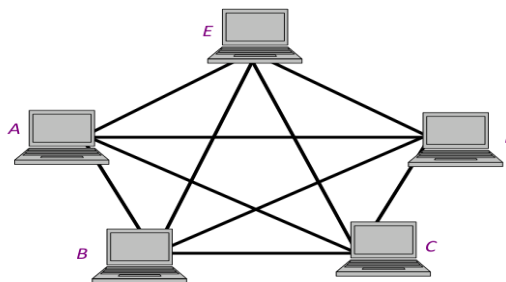
Types of connections

- Point-to-point: Dedicated link between two devices.
- Multipoint (multidrop): More than two devices share a single link.

Physical topology: The physical layout of a network and the geometric relationship of its links/linking devices.

Four basic topologies: Mesh, Star, Bus and Ring.

Mesh Topology



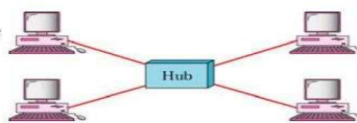
- In mesh topology every device has a dedicated point to point link to every other device
- We need $n(n-1)$ links
- Communication is duplex mode
- Every device on the network must have $n-1$ i/o ports to connect to the other $n-1$ devices.

Mesh: Every device has a dedicated point-to-point link to every other device. Pros: dedicated links, robustness, privacy and easy fault identification. Cons: much cabling and I/O ports; difficult to move.

For n devices, the PPT states that $n(n-1)$ links are needed and each device needs $n-1$ I/O ports.

Star Topology

- Each device has a dedicated point-to-point link only to a central controller, usually called a hub.
- No direct traffic and link between devices
- Advantages
 - Less expensive
 - Easy to install and reconfigure
 - Robustness
- Disadvantage
 - Single point of failure
- More cabling is required than other topologies

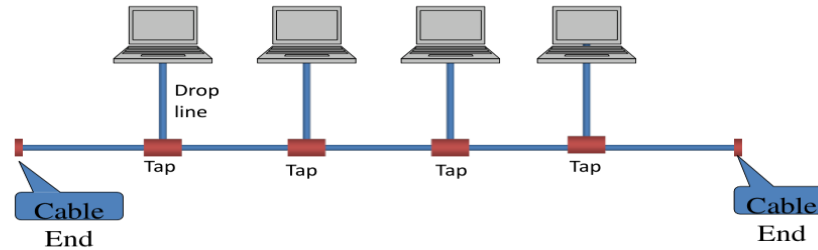


Star: Each device has a dedicated point-to-point link to a central controller, usually a hub. It is easy to install/reconfigure; the central device is a single point of failure.

PPT reference: Slides 16–24.

6. Explain Bus and Ring Topologies.

Bus Topology

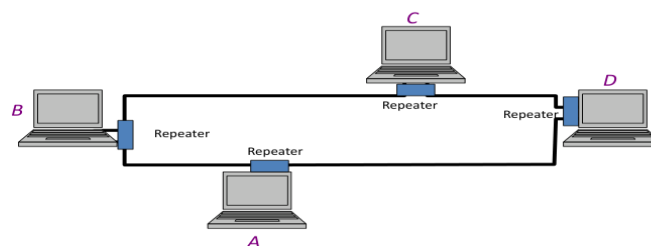


26

Bus topology: A multipoint topology in which all devices are linked through a backbone cable. Nodes connect through drop lines and taps.

- Advantage: Easy installation.
- Disadvantages: Difficult reconnection/fault isolation; a broken/faulty bus cable stops all transmission.

Ring Topology



27

Ring topology: Each device has a dedicated point-to-point connection with the two devices on either side. A signal passes around the ring until it reaches its destination; each device incorporates a repeater.

- Pros: Easy to install and identify faults.
- Cons: Delay in a large ring; a break in the ring can stop all transmission.

PPT reference: Slides 25-28.

7. Explain Network Types: LAN, MAN and WAN.

Type	Definition / coverage	Key point
LAN	Privately owned network in a single office, building or campus.	Each host has an address; packets carry source and destination host addresses.
MAN	Metropolitan-scale network type, distinguished by size/geographical coverage/ownership.	Intermediate geographical scope between LAN and WAN.
WAN	Wider coverage: town, state, country or even the world.	Interconnects connecting devices; can be point-to-point or switched.

WAN examples from the PPT

- Point-to-point WAN: Two communicating devices are connected.
- Switched WAN: More than two ends; combines point-to-point WAN links through switches.

PPT reference: Slides 30-34.

8. Connection-Oriented vs Connectionless Service.

Connection-oriented: Modeled after the telephone system — establish a connection → use it → release it.

Connectionless: Modeled after the postal system — each message carries the full destination address and is routed independently.

Criterion	Connection-Oriented	Connectionless
Connection	Prior connection is established.	No prior connection.
Resources	Resources may be associated with the connection.	No prior allocation required.
Reliability	Can provide reliable transfer.	Reliability is not guaranteed; best effort.
Transfer	Can use circuit switching or packet switching.	Implemented using packet switching.
Suitability	Long and steady communication.	Bursty transmissions.
Packet travel	Established path.	Packets may travel independently.

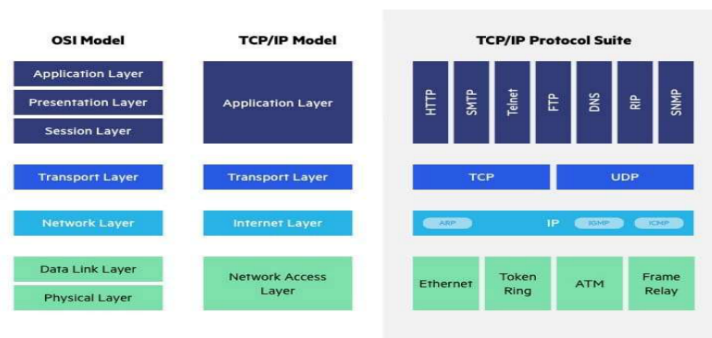
Reliability note: Reliable service can use acknowledgements, but acknowledgements add overhead and delay.

PPT reference: Slides 39–42.

9. Explain Network Devices: Modem, Hub, Switch and Router.

Device	Function
Modem	Modulates/demodulates or converts signals between forms. Internet modems receive signals from an ISP and convert them into a usable form; they commonly connect to a router.
Hub	Joins multiple devices on the same LAN and broadcasts a received message from one port to all remaining ports without examining the frame.
Switch	Examines the incoming frame's MAC address and sends data to the matching destination port. Maintains a MAC-address table.
Router	Directs data from one network to another by examining destination IP addresses and using routing tables/protocols.

OSI versus TCP/IP



PPT reference: Slides 43-50. The PPT also lists repeater, bridge, gateway, router and NIC as network devices.

10. Explain the OSI Model.

OSI (Open System Interconnection) is a layered framework for designing network systems. It consists of seven separate but related layers.

The OSI 7 Layers

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

60

Layer functions

- 7 Application: User interfaces and services such as electronic mail and remote file access/transfer.
- 6 Presentation: Translation, encryption and compression.
- 5 Session: Establishes, maintains and synchronizes interaction; dialog control and synchronization.
- 4 Transport: Process-to-process delivery, segmentation/reassembly, end-to-end flow and error control.
- 3 Network: Source-to-destination delivery, logical addressing and routing.
- 2 Data Link: Hop-to-hop delivery, framing, physical addressing, flow/error/access control.
- 1 Physical: Interface with the medium, bit representation, data rate, synchronization, line configuration, topology and transmission mode.

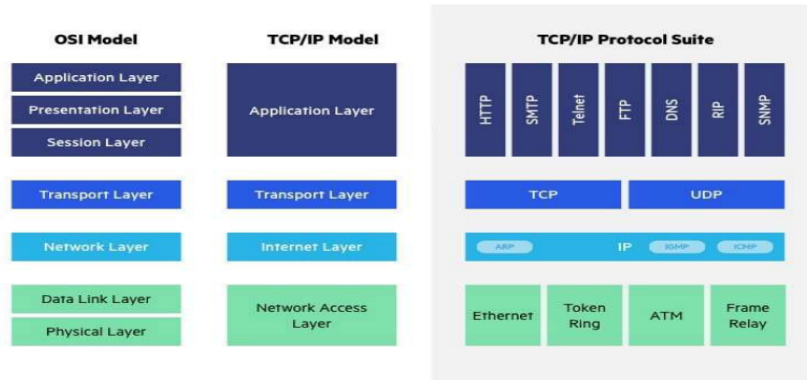
Remember: Physical → bits; Data Link → frames; Network → packets; Transport → segments; Application → user services.

PPT reference: Slides 52-82.

11. Explain TCP/IP Protocol Suite and OSI vs TCP/IP.

TCP/IP is also called the Internet Protocol Suite. It consists of communication protocols used for the Internet and similar networks. Two important protocols are TCP and IP.

OSI versus TCP/IP



TCP/IP layer	OSI correspondence	Important point
Application	Session + Presentation + Application	Combines the functions of three OSI layers.
Transport	Transport	Includes TCP and UDP.
Internet	Network	IP sends datagrams based on address; it is unreliable, connectionless and best effort.
Host-to-network	Physical + Data Link	Supports physical/data-link functions.

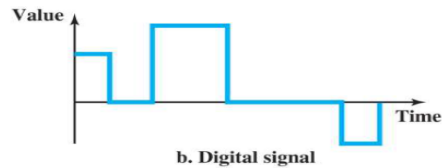
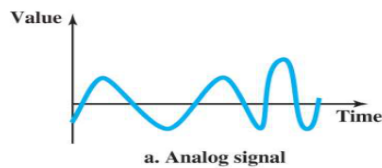
OSI vs TCP/IP: Session and Presentation are not separate layers in TCP/IP; their functions are generally handled within the Application layer.

PPT reference: Slides 83-90.

12. Explain Analog and Digital Signals.

SIGNALS

- **Signals** can be either analog or digital.
- An **analog signal** has infinitely many levels of intensity over a period of time.
- As the wave moves from value *A* to value *B*, it passes through and includes an infinite number of values along its path.
- A **digital signal**, on the other hand, can have only a limited number of defined values.
- Although each value can be any number, it is often as simple as 1 and 0.



Analog	Digital
Can take infinitely many intensity levels.	Has a limited number of defined values.
Sine wave is described by peak amplitude, frequency and phase.	Often represented using 1 and 0; may use more than two levels.
Normally periodic analog signals are used.	Normally nonperiodic digital signals are used.

Bit rate: Number of bits sent in one second.

Bit length: Distance occupied by one bit on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

PPT reference: Slides 9–23 of the Physical Layer section.

13. Signal Impairment — Causes.

SIGNAL IMPAIRMENT

- Signals travel through transmission media, which are not perfect.
- The imperfection causes signal impairment.
- This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.
- **“What is sent is not what is received”.**
- Three causes of impairment are **attenuation**, **distortion**, and **noise**.

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36

Signals travel through media that are not perfect, so the signal at the end of the medium is not the same as the signal at the beginning.

Three causes: Attenuation + Distortion + Noise

1. Attenuation: Signal loses energy while travelling; amplification is needed to compensate.

$$\text{dB} = 10 \log_{10}(P_2 / P_1)$$

Negative dB → attenuation; positive dB → amplification.

2. Distortion: Signal components can experience different delays/phases, changing the received signal shape.

3. Noise:

- Thermal noise: Random motion of electrons creates an extra unwanted signal.
- Induced noise: Motors/appliances act as sending antennas and the medium acts as the receiving antenna.
- Crosstalk: Effect of one wire on another; one wire acts as a sending antenna and the other as a receiving antenna.
- Impulse noise: A high-energy spike in a very short time, such as from power lines or lightning.

PPT reference: Slides 36–44.

14. Data Rate Limits — Nyquist and Shannon.

Data rate depends on bandwidth available, signal levels used and quality/noise level of the channel.

Nyquist — noiseless channel

$$\text{Bit Rate} = 2 \times \text{Bandwidth} \times \log_2 L$$

Here L is the number of signal levels.

PPT example: $B = 3000 \text{ Hz}$, $L = 2 \rightarrow \text{Bit Rate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$.

Increasing the number of levels can increase the theoretical rate, but makes it harder for the receiver to distinguish levels and reduces reliability.

Shannon — noisy channel

$$\text{Capacity} = \text{Bandwidth} \times \log_2(1 + \text{SNR})$$

High SNR $\rightarrow$ less corruption by noise. Low SNR $\rightarrow$ more corruption.

Important: Shannon capacity is a characteristic of the channel, not the transmission method. Increasing signal levels cannot exceed the channel capacity.

PPT reference: Slides 45-56.

15. Performance: Bandwidth, Throughput, Latency, BDP and Jitter.

Performance

- Up to now, we have discussed the tools of transmitting data (signals) over a network and how the data behave.
- One important issue in networking is the performance of the network—how good is it?

Bandwidth

- One characteristic that measure network performance is bandwidth.

In networking, we use the term *bandwidth* in two contexts.

- ☐ The first, *bandwidth in hertz*, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- ☐ The second, *bandwidth in bits per second*, refers to the speed of bit transmission in a channel or link.

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58

Metric	Meaning
Bandwidth	A measure/characteristic of a link indicating available frequency range or potential carrying capacity.
Throughput	How fast data can actually be sent. If bandwidth is B and throughput is T, typically $T < B$.
Latency / Delay	Time taken for data to travel through the network; transmission, propagation, processing and queuing contribute to delay.
Bandwidth-delay product	Bandwidth $\times$ delay; defines the number of bits that can fill the link.
Jitter	Variation in packet delays; important for time-sensitive audio/video.

Transmission time = Message size / Bandwidth

Bandwidth-delay product = Bandwidth $\times$ Delay

PPT example: 12,000 frames/minute $\times$ 10,000 bits/frame on a 10-Mbps network $\rightarrow$ throughput = 2 Mbps.

Jitter example: packet delays of 20 ms, 45 ms and 40 ms cause jitter for a real-time application.

PPT reference: Slides 57–72.

16. Digital-to-Digital Conversion and Line Coding.

Digital-to-digital conversion represents digital data using digital signals. The PPT identifies line coding, block coding and scrambling; line coding is always needed.

Unipolar scheme

Unipolar Scheme

- In a unipolar scheme, all the signal levels are on one side of the time axis, either above or below.
- **NRZ (Non-Return-to-Zero)**
- Traditionally, a unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the positive voltage defines bit 1 and the zero voltage defines bit 0.
- It is called NRZ because the signal does not return to zero at the middle of the bit.

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In the PPT example, 1 can be represented by a positive voltage and 0 by zero voltage.

Polar schemes

Polar Schemes

- In the first variation, NRZ-L (NRZ-Level), the level of the voltage determines the value of the bit.
- In the second variation, NRZ-I (NRZ-Invert), the change or lack of change in the level of the voltage determines the value of the bit.
- If there is no change, the bit is 0; if there is a change, the bit is 1.

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NRZ-L: voltage level determines the bit value.

NRZ-I: change/lack of change determines the bit. No change → 0; change → 1.

PPT reference: Slides 74–86.

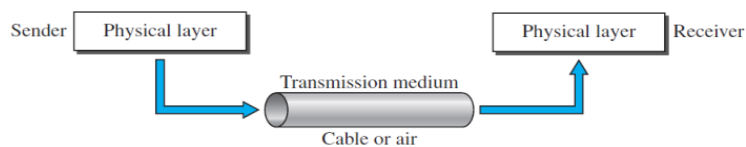
17. Compare NRZ-L, NRZ-I, RZ and Manchester / Differential Manchester.

Scheme	Main idea	Important point
NRZ-L	Voltage level determines bit.	Baseline wandering and synchronization problems occur, especially with long runs.
NRZ-I	Transition/no transition determines bit: change = 1, no change = 0.	Same problems remain, but are less severe than NRZ-L according to the PPT.
RZ	Uses positive, negative and zero levels; returns to zero during the bit.	Helps address NRZ clock synchronization problem.
Manchester	Combines RZ mid-bit transition with NRZ-L.	Always a mid-bit transition, providing synchronization.
Differential Manchester	Combines RZ and NRZ-I.	Always a mid-bit transition; value is determined by transition/no transition at the beginning.

Transmission Media

- Transmission media are actually located below the physical layer and are directly controlled by the physical layer.
- We could say that transmission media belong to layer zero.
- Figure shows the position of transmission media in relation to the physical layer.

Figure 7.1 *Transmission medium and physical layer*



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92

Exam focus: Level → NRZ-L; transition/no transition → NRZ-I; mid-bit transition → Manchester.

PPT reference: Slides 83–90.

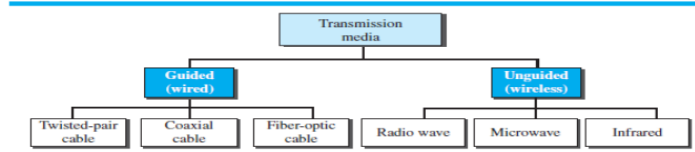
18. Explain Transmission Media — Guided and Unguided.

A transmission medium carries information from source to destination. In data communications, the medium is usually free space, metallic cable or fiber-optic cable.

Transmission Media

- In telecommunications, transmission media can be divided into two broad categories: **guided and unguided**.
- **Guided media** include twisted-pair cable, coaxial cable, and fiber-optic cable.
- **Unguided medium** is free space.

Figure 7.2 Classes of transmission media



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95

Unguided Media: Wireless

- **Unguided medium** transport electromagnetic waves without using a physical conductor.
- This type of communication is often referred to as **wireless communication**.
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

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96

Guided media

- **Twisted pair:** Two conductors twisted together. Twisting helps balance noise/crosstalk. UTP is common; STP adds metallic shielding but is bulkier and more expensive.
- **Coaxial cable:** Central copper conductor, insulation and outer metallic conductor/shield. It carries higher frequency ranges than twisted pair and provides shielding.
- **Fiber-optic cable:** Glass/plastic medium carrying signals as light; high bandwidth, low attenuation and immunity to electromagnetic interference. Installation/maintenance require expertise and interfaces are relatively expensive.

Unguided: Free space; the signal propagates through air/free space.

PPT reference: Slides 93–134. The handwritten list asks for guided & unguided; the PPT gives the detailed treatment mainly for guided media.

19. Explain Switching and Packet Switching.

A switched network consists of interlinked nodes called switches. Switching creates temporary connections between devices and is more practical than direct connections between every pair in a large network.

Packet switching

Packet Switching

1. Datagram Networks

In a **datagram network**, each packet is treated independently of all others. Even if a packet is part of a multipacket transmission, the network treats it as though it existed alone. Packets in this approach are referred to as **datagrams**.

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A message sent through a packet-switched network is divided into packets of fixed or variable size. Packet size is determined by the network and governing protocol.

- No reserved bandwidth or scheduled processing time is allocated for each packet.
- Resources are allocated on demand, generally first-come, first-served.
- Two types: datagram networks and virtual circuit networks.
- Datagram network: each packet is treated independently; packets of one message may travel different paths.
- Datagram networks are also called connectionless networks; there are no setup or teardown phases.

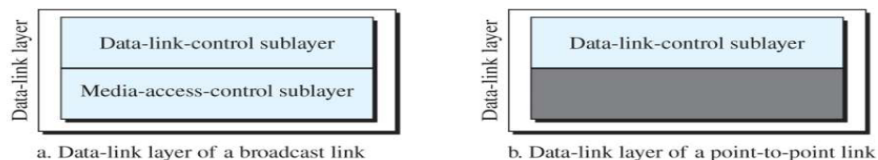
Exam line: Packet switching divides a message into packets and shares network resources on demand.

PPT reference: Switching slides 135–139 and the packet-switching section.

20. Explain the Sublayers of the Data-Link Layer.

Two Sublayers

- To better understand the functionality of and the services provided by the link layer,
- we can divide the data-link layer into two sublayers: **data link control (DLC) and media access control (MAC)**.



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18

The data-link layer can be divided into two sublayers.

Sublayer	Role
DLC	Deals with issues common to point-to-point and broadcast links. Functions include framing, flow control and error control.
MAC	Deals with issues specific to broadcast/shared links and coordinates access to a common medium.

Data-link functions

- Framing: Divides the bit stream into manageable frames.
- Physical addressing: Adds sender and receiver addressing information.
- Flow control: Controls data flow to avoid bottlenecks.
- Error control: Helps avoid data loss/corruption.
- Access control: Determines which device has control over the transmission line.

Key line: Data-link layer = hop-to-hop delivery of frames.

PPT reference: Slides 18-22 of the Data-Link Layer section.

21. Explain Character-Oriented and Bit-Oriented Framing.

Framing: Packs bits into frames so that each frame is distinguishable. Variable-size framing uses character-oriented and bit-oriented approaches.

Character-oriented / byte-oriented

Character-Oriented or Byte oriented Framing

- In this type of framing, data to be carried are 8-bit characters from a coding system such as ASCII.
- The header, normally carries the source and destination addresses and other control information.
- The trailer, carries error detection redundant bits are also multiples of 8 bits.
- To separate one frame from the next, an 8-bit (1-byte) **flag** is added at the beginning and the end of a frame.

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- Data is carried as 8-bit characters, such as ASCII.
- Header normally carries source/destination addresses and control information.
- Trailer carries error-detection redundant bits.
- An 8-bit flag is placed at the beginning and end of the frame.
- Byte stuffing: if a flag-like character occurs inside the data, an ESC character is inserted; the receiver removes it and treats the next character as data.

Bit-oriented framing

Bit-Oriented Framing

- In bit-oriented framing, data is a sequence of bits.
- To separate one frame from another, we normally use an **8-bit** pattern **flag (01111110)** as the delimiter to define the beginning and the end of the frame, as shown in the next slide.
- This flag can create the same type of problem we saw in the character-oriented protocols.
- That is, if the flag pattern appears in the data, we need to somehow inform the receiver that this is not the end of the frame.

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The normal 8-bit flag is 01111110. If the flag pattern could appear in the data, bit stuffing is used.

PPT reference: Slides 23–34.

22. Explain Bit Stuffing and Unstuffing.

In bit-oriented framing, the flag pattern is 01111110. If a similar pattern occurs in the data, it may be mistaken for the end of the frame.

Rule: When a 0 followed by five consecutive 1 bits is encountered, an extra 0 is added.

The receiver removes the extra stuffed 0 during unstuffing.

Example

Before: 0 1 1 1 1 1

After stuffing: 0 1 1 1 1 0 1

Why? The stuffed bit prevents the flag sequence from inadvertently appearing in the frame. The receiver removes the extra bit to recover the original data.

PPT reference: Slides 31-34.

23. Explain Types of Errors and Redundancy.

Error type	Meaning
Single-bit error	Only one bit of the data unit changes from 1→0 or 0→1.
Burst error	Two or more bits in the data unit change from 1→0 or 0→1.

Redundancy

Redundancy is the central concept in error detection/correction. Extra bits are sent with the data; the sender adds them and the receiver uses/removes them after detecting or correcting errors.

Block coding: Divide the message into blocks of k data bits (datawords), add r redundant bits, and obtain $n = k + r$ bits called a codeword.

Minimum Hamming distance: To guarantee detection of up to s errors, $d_{\min} = s + 1$.

PPT reference: Slides 35–54.

24. Explain Parity-Check Code.

Table:2 Simple parity-check code C(5, 4)

<i>Datawords</i>	Codewords	<i>Datawords</i>	Codewords
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

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56

Parity-check code is a familiar error-detecting code and a linear block code.

A k-bit dataword becomes an n-bit codeword, where $n = k + 1$. The extra bit is the parity bit.

Even parity: Select the parity bit so the total number of 1s in the codeword is even.

For a 4-bit dataword $a_0a_1a_2a_3$, the PPT gives:

$$r_0 = a_0 + a_1 + a_2 + a_3 \text{ (modulo-2)}$$

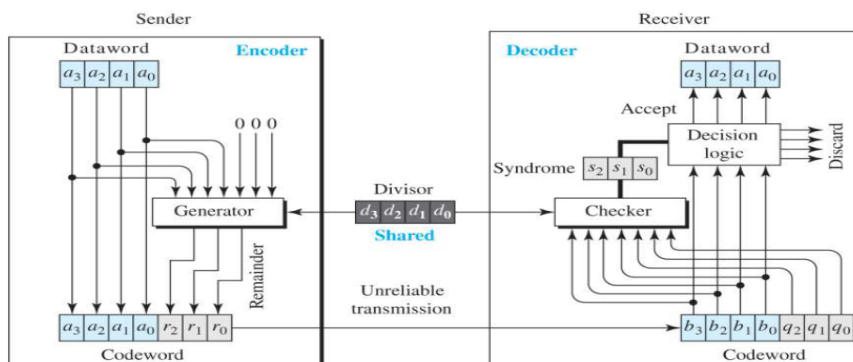
Example: Dataword = 1011 has three 1s, so parity bit = 1. Codeword = 10111, giving four 1s.

Detection: It is guaranteed to detect one single-bit error and can detect any odd number of errors. It cannot detect an even number of errors in general.

PPT reference: Slides 55–65.

25. Explain Cyclic Redundancy Check (CRC).

CRC encoder and decoder



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69

CRC is a subset of cyclic codes and is used in networks such as LANs and WANs. In a cyclic code, a cyclic shift of a codeword produces another codeword.

CRC Encoder — steps

- Dataword has k bits; final codeword has n bits.
- Append $n - k$ zeros to the right side of the dataword.
- Feed the augmented dataword into the generator/divisor.
- Perform modulo-2 division.
- Discard the quotient and append the remainder to the original dataword to form the codeword.

CRC Decoder

- Feed the received n -bit codeword to a checker that is a replica of the generator.
- The resulting remainder is the syndrome.
- All-zero syndrome → accept the dataword as having no detected error.
- Non-zero syndrome → error; discard the codeword/dataword.

Standard polynomials shown in the PPT: CRC-8, CRC-10, CRC-16 and CRC-32. CRC-16 is listed for HDLC and CRC-32 for LANs.

Exam procedure: Append zeros → modulo-2/XOR division → take remainder → append remainder → divide the received codeword again → zero remainder means accept.

PPT reference: Slides 66–75.

LAST-MINUTE REVISION SHEET

Definitions

- Data communication = exchange of data between two devices through a transmission medium.
- Network = interconnection of devices capable of communication.
- Point-to-point = dedicated link; multipoint = shared link.
- OSI = seven-layer model.
- TCP/IP = Internet Protocol Suite; original four-layer view in the PPT.
- Signal impairment = attenuation + distortion + noise.
- Throughput = actual rate; bandwidth = potential link capacity.
- Jitter = variation in packet delay.
- Packet switching = message divided into packets; resources allocated on demand.
- DLC = framing + flow/error control; MAC = shared/broadcast access.
- Bit stuffing = insert 0 after five consecutive 1s following a 0.
- Parity = add one bit so total 1s are even.
- CRC = modulo-2 division; remainder becomes redundant bits.

Formulas

Nyquist: Bit Rate = $2B \log_2 L$
Shannon: Capacity = $B \log_2(1 + \text{SNR})$
Transmission time = Message size / Bandwidth
Bit length = Propagation speed $\times$ Bit duration
dB = $10 \log_{10}(P_2/P_1)$
Detect up to s errors: $d_{\min} = s + 1$

Source: Uploaded Unit-I Computer Network and Communication PPT, Course Code CS3001-1. The handwritten important-question list was used to select the topics.